# Introduction to Classification
# (IART Lab Exercises)

April 26, 2026

## Exercise 1

Use Decision Trees to predict which of a set of prospects (potential customers) are buying a car.

### a) Load development/training data

Load the sheet "car sales-historical data" that is available in the excel file provided.

### b) Learn a decision tree

The target variable is "Bought?".

After learning, visualize and analyze it.

### c) Load new data

Load the sheet "car sales-prospects" that is available in the excel file provided.

## Exercise 2

Develop a Decision Trees to predict which customer is buying a car or not, including an evaluation of its predictive performance.

### a) Load development/training data

Load the sheet "car sales-historical data" that is available in the excel file provided.

### b) Split the data into a training and a test set

Use 70% for training and the rest for testing.

### c) Learn a decision tree from the training data

The target variable is "Bought?".

After learning, visualize and analyze it. Is it different from the previous one?

### d) Evaluate the decision tree on the test data

First, use the decision tree to make the predictions and then evaluate those predictions.

## Exercise 3

Compare the model developed to the majority class baseline.

Is the model useful?

### a) Compute the majority class

Count the number of examples from each class and then select the highest value.

### b) Compute the default accuracy

Add a column to the predictions with the majority class on all rows and evaluate it as a prediction.

## Exercise 4

Compare the test error with the training error.

Is the model developed overfitting?

### a) Compute the training error

Evaluate the predictions made by the model on the data used for training it.

## Exercise 5

Change the train/test split of the data and verify the effect on predictive performance.

### a) Change the seed of the pseudo-random number generator of the partitioning operator

Compare the training and test errors obtained on different partitions.

## Exercise 6

Repeat the exercises for the data in the "credit" sheet of the same excel file.

You may have to adjust the pipeline due to differences in this dataset relative to the first one.

## Exercise 7

Repeat the exercises for the data in the "telemarketing" sheet of the same excel file.

You may have to adjust the pipeline due to differences in this dataset relative to the first one. Analyze the data to identify what the issue is.